

Superkeys

Superkey

- A set of attributes such that …
- If tuples have the same superkey values,
- Then tuples are the same.



Find the superkeys

Student (fname, lname, netid, classification, phone, birthdate)

Find the superkeys

Teaches (instructor, course)

Finding superkeys correctly

Need to consider semantics of schema, not just the data in a single instance.

However, you might be given just the data, and you need to infer the intended semantics.



Are relations guaranteed to have a superkey?

Superkey definition

A set K of attributes is a superkey of R iff

- For every instance R' of R,
- For all tuples t_1 and t_2 in R',
- If $t_1[K] = t_2[K]$, then $t_1 = t_2$.